

OPTION TRADE TYPE ②: TWO OR MORE UNDERLYINGS

The risk in the trade described above is that the volatility actually realized may be greater or less than the volatility we anticipated. As a result, option trades of type ② with a single underlying can be seen as fundamental bets on whether particular macroeconomic scenarios materialize. As shown in Table 3.1, macroeconomic events tend to impact both the direction of the underlying swap rates and the realized volatility level, resulting in a significant correlation between them. Though option trades of type ② have no direct exposure to the underlying due to delta hedging, their exposure to the level of realized volatility does link them to the corresponding macroeconomic environment of high or low uncertainty (as caused by such events as central bank action, banking crises, etc.).

In order to reduce the macroeconomic exposure, the idea is to combine two of those trades, a long and a short option (both delta hedged), thus to hedge against the *overall* level of future realized volatility and to leave exposure just to the *relative* realized volatility distribution across the yield curve. This leads to relative value positions exploiting structural mismatches along the implied and realized volatility curves.

For example, buying 2Y straddles, selling 10Y straddles, and delta hedging both is a *box trade* between the current implied volatility curve and the future realized volatility curve, which will profit if the future realized volatility curve is flatter than the current implied volatility curve. If hedge ratios are constructed properly, the overall level of future realized volatility will have no impact on that position.

In order to screen the market for these trading opportunities, an analyst could look at a graph comparing the current implied and realized volatility curves (Figure 19.17).

In this example, we can first observe that implied volatility is currently above the recent realized volatility for all underlying maturities, but with a minimum difference for 5Y. This means that trades of type ② with a single underlying (i.e. a long delta hedged option position exploiting the relative cheapness of current implied volatility versus the average realized volatility¹⁶) are best expressed in the 5Y segment. Of course, the risk to these positions is that the realized volatility until expiry will turn out to be below the current implied volatility. While Figure 19.13 suggests that this is rather unlikely,

¹⁶Though current implied volatility is slightly rich versus *current* realized volatility (see Figure 19.13).

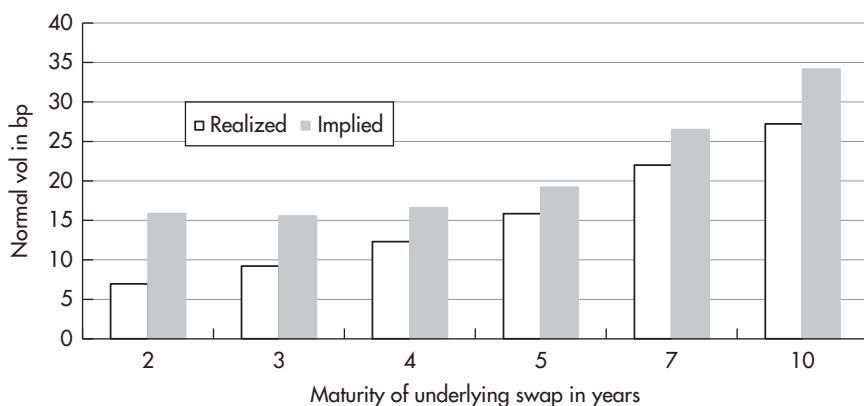


FIGURE 19.17 Current realized and implied volatility across the JPY curve.

Sources: data – Bloomberg; chart – Authors.

Data period: “Current” market data as of 18 Jun 2012.

it does expose a long option trade of type ② to the macroeconomic risk of an environment that produces extremely low realized volatility, such as BoJ announcements removing all uncertainty about future interest rates. Above, we have developed tools to assess the risk/return profiles of these positions.

Now we look at the same information displayed in Figure 19.17 differently, with the eyes of a relative value analyst who wants to be hedged against macroeconomic impacts as well as possible and just to exploit relative mismatches in the options market. Thus, our focus shifts from the *level* of implied versus realized volatility to their distribution over different maturities, to the *shape* of the implied volatility curve relative to the realized volatility curve. And it jumps out at the “relative value eye” immediately that the realized volatility curve is much steeper than the implied volatility curve between 2Y and 5Y. If we have reason to believe the realized volatility curve will remain steep in future as well, we could exploit the mismatch between the steepness of the implied curve relative to the realized volatility curve by buying 3M5Y straddles versus selling 3M2Y straddles and delta hedging both. Then, an increase in the overall level of realized volatility would cause the delta hedging of the long straddle position to win as much as the delta hedging of the short straddle position loses (if hedge ratios are appropriate). Thus, we

are hedged against changes to the overall level of realized volatility. Instead, we are exposed to the difference between realized and implied volatility in 5Y *relative* to that difference in 2Y.

In order to assess that exposure, we recommend adjusting Figure 19.13 to two underlyings and thus depicting the history of the realized volatility *difference* between 5Y and 2Y versus the current implied volatility *difference* between 5Y and 2Y. Figure 19.18 shows the historical evolution of the 2Y-5Y realized volatility curve steepness versus the current 2Y-5Y implied volatility curve steepness.

In this case, the current implied volatility curve steepness was never matched by the realized volatility curve steepness during the past five years. Thus, if we had entered the long 5Y short 2Y straddle position at current implied volatility levels and delta hedged it until expiry at any point during the past five years, we would always have made money (on average 10 bp). Of course, it is possible that the future 5Y-2Y realized volatility spread will be below the current implied volatility spread. However, Figure 19.18 makes us confident that this is unlikely, given that the range of the 5Y-2Y realized volatility spread has over the tumultuous past five years been stably in a range between 6 bp and 27 bp, and even the lower end of that range is still away

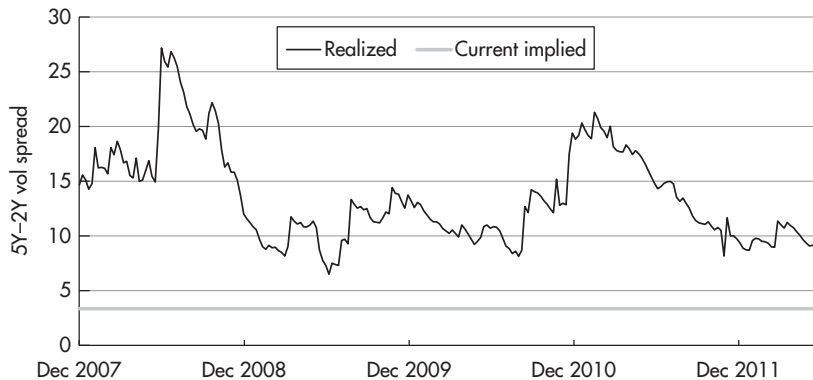


FIGURE 19.18 History of 2Y-5Y JPY realized volatility curve steepness versus current 2Y-5Y JPY implied volatility curve steepness.

Sources: data – Bloomberg; chart – Authors.

Data period: 24 Dec 2007 to 18 Jun 2012, weekly data; “current” market data as of 18 Jun 2012.

from the current implied volatility spread of 3 bp. Moreover, given recent BoJ announcements, we have no reason to expect realized volatility at the short end of the yield curve to increase relative to realized volatility in 5Y as much as is priced into the implied volatility curve. Thus, we have not only a statistical reason (history of Figure 19.18) but also a fundamental reason to believe that the steepness of the realized volatility curve will continue in the future and therefore for the current flatness of the implied volatility curve to be a trading opportunity.

As always, relative value considerations can also be used for asset selection purposes when expressing fundamental views. In this case, an analyst wanting to position for the overall future realized volatility level to be above the current implied volatility level (see Figure 19.13) could find best value in choosing long 5Y straddles to express his view (by looking at Figure 19.17).

Comparing Figure 19.13 and Figure 19.18, we observe significant correlation (0.71) between the level of realized volatility and the slope of the 2Y-5Y realized volatility curve: when overall volatility increases, the volatility curve tends to steepen. Depending on the goal, one can react to that observation in two ways:

- If we want to express the fundamental view of a high future realized volatility, we could do so via a 2Y-5Y straddle spread rather than a long 5Y straddle (trade type ② with two rather than one underlyings). If our fundamental view is correct, both the 5Y straddle and the 2Y-5Y straddle spread should perform well, as the two positions are highly correlated with one another. But the 2Y-5Y straddle spread is preferred, as it has a much better risk/return profile. While the 2Y-5Y straddle spread offers a 5 bp cushion against the volatility curve flattening associated with lower overall realized volatility (Figure 19.18), a long 5Y straddle will only win if realized volatility increases by at least 3 bp from its current level (Figure 19.13). Thus, for this goal we exploit the correlation between the “fundamental” and the “relative value” trade to improve the P&L from the same macroeconomic events.
- If, on the other hand, our goal is to hedge against the impacts of the overall realized volatility level, we can adjust the hedge ratio of the 2Y-5Y straddle spread in order to immunize against the exposure to the “direction” of realized volatility. Selling 1.23 2Y straddles for every 5Y straddle bought results in neutrality against the overall level of realized volatility. Figure 19.19 shows the history of the non-directional 2Y-5Y realized volatility curve steepness and illustrates the attractiveness of the pure relative value trade.

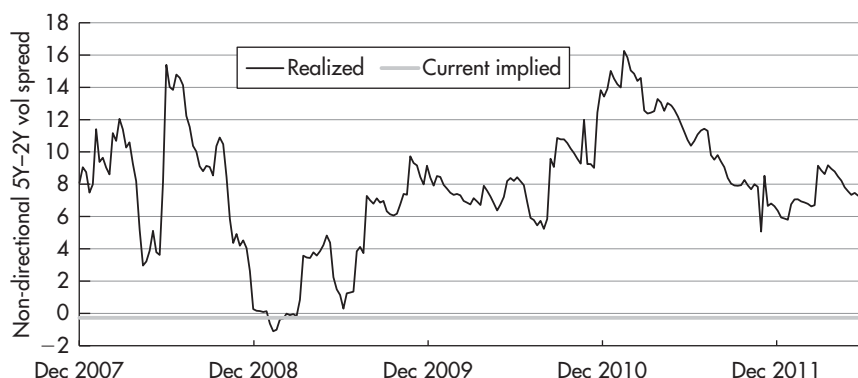


FIGURE 19.19 History of non-directional 2Y-5Y JPY realized volatility curve steepness versus current 2Y-5Y JPY implied volatility curve steepness.

Sources: data – Bloomberg; chart – Authors.

Data period: 24 Dec 2007 to 18 Jun 2012, weekly data; “current” market data as of 18 Jun 2012.

Note that option trade types ① and ② both depend on the difference between current implied volatility and future realized volatility. Therefore, both look for opportunities in where the current market pricing (implied volatility) is out of line with the expected realized volatility. However, since the way that the two types of option trades exploit these mismatches is different, we recommend displaying the same information (implied versus realized volatility) in two different ways: for type ① in terms of premium versus directionality (Figure 19.11) and for type ② in terms of implied versus realized volatility curves (Figure 19.17).

OPTION TRADE TYPE ③: FACTOR MODEL FOR THE VEGA SECTOR

By contrast, straddles in the vega sector of the volatility surface are almost unaffected by realized volatility in the underlying and purely exposed to changes in the implied volatility level. This allows analysis of implied volatility in abstraction from its link to external variables, focusing on the internal relationships between different points on the implied volatility surface by treating them as purely statistical time series (without making use of the knowledge that the time series represents option volatilities, which is a market consensus about future realized volatility). In brief, type ③ strategies

are option positions that trade options not *as* options but as abstract time series, not linked via delta hedging to an external variable.

Consequently, while the analysis tools for types ① and ② were based on the structural link of options (*as* such, as an option *on something*) to the underlying (i.e. on the link between implied and realized volatility), analysis tools for type ③ need to consider options in abstraction from that link and focus on the statistical relationships between different points on the implied volatility surface. An equivalent statement is that option trades of type ② are analyzed within the Black–Scholes framework,¹⁷ while type ③ is conceptually different and treats options in abstraction from their link through delta hedging to the underlying. As in Chapter 13, where we analyzed the statistical properties of credit default swaps (CDS) in abstraction from a default situation, we now consider the mechanisms in the vega sector of the volatility surface in abstraction from the connection (through delta hedging) to the realized volatility.

Thus, the vega sector of the volatility surface is the right place to apply a statistical tool such as a PCA-based factor model. And conversely, the application of PCA on the volatility surface confirms empirically its differentiation into gamma and vega sectors. Running a PCA on the whole volatility surface reflects the break between the gamma and vega segments in different sensitivities to the first factor. As Figure 19.20 shows, sensitivity starts decreasing as the expiry increases – and even turns negative for very long expiries. The fact that the first eigenvector has entries with different signs is a clear indication for segmentation in the input variables and justifies empirically our approach to limit the statistical analysis to the vega sector.

Restricting the data input to options with at least 2Y to expiry indeed solves the problem and returns a first eigenvector which has only negative entries (Figure 19.21). That is, this first factor can be interpreted as the overall implied volatility level, which affects all instruments in the same direction, though to a different extent. Figure 19.21 also reveals the location of the break between the gamma and vega sectors in the JPY swaption market. For the 2Y expiry, only underlyings of more than 5Y are clearly in the vega segment, while, for a 5Y expiry, all options can be considered to be in the vega sector, independent of the maturity of the underlying swap. In terms of Figure 19.5, this means that, in the case of Japan, the borderline between the two segments is somewhat tilted to the right. That is, the longer the maturity of the underlying swap, the shorter the expiry needs to be in order for an option to be part of the vega sector (e.g. four years for an option on 1Y swaps, three years for an option on 2Y swaps, and two years for an option on 5Y swaps).

¹⁷ And trades of type ① can be considered in that framework as one-period (instead of continuous) delta hedging strategies.

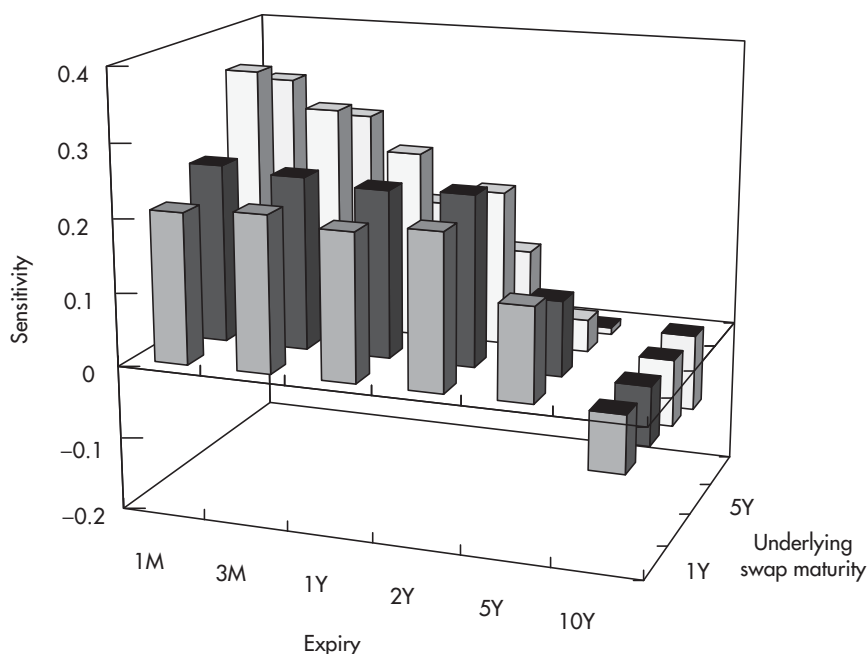


FIGURE 19.20 First eigenvector of a PCA on the whole JPY volatility surface.

Sources: data – Bloomberg; chart – Authors.

Data period: 5 Jan 2009 to 19 Sep 2011, weekly data.

Now that PCA has revealed the sector of the volatility surface on which it is applicable, we can run through the usual analytical process, whose results are depicted in Figure 19.22, Figure 19.23, and Figure 19.24.

Factor 1 can be interpreted as the overall level of implied volatility, with the shape of the first eigenvector showing the way changes in overall volatility impact the volatility surface. Factor 2 represents the steepness of volatility curves (same underlying, different expiries). If factor 2 increases, options with a long expiry increase relative to those with a short expiry, almost independent of the underlying swap maturity. Factor 3 represents the steepness of volatility curves (same expiry, different underlyings). If factor 3 increases, options with a long underlying swap maturity increase relative to those with a short underlying swap maturity, almost independent of the expiry.

Thus, PCA can be considered a decomposition of the volatility surface into its two dimensions, with the second and third factors quantifying the variation that occurs in a specific dimension. Factor 1 represents the *overall* implied volatility level and correspondingly affects *both* dimensions. Factors 2 and 3 explain that part of implied volatility which occurs in a specific dimension

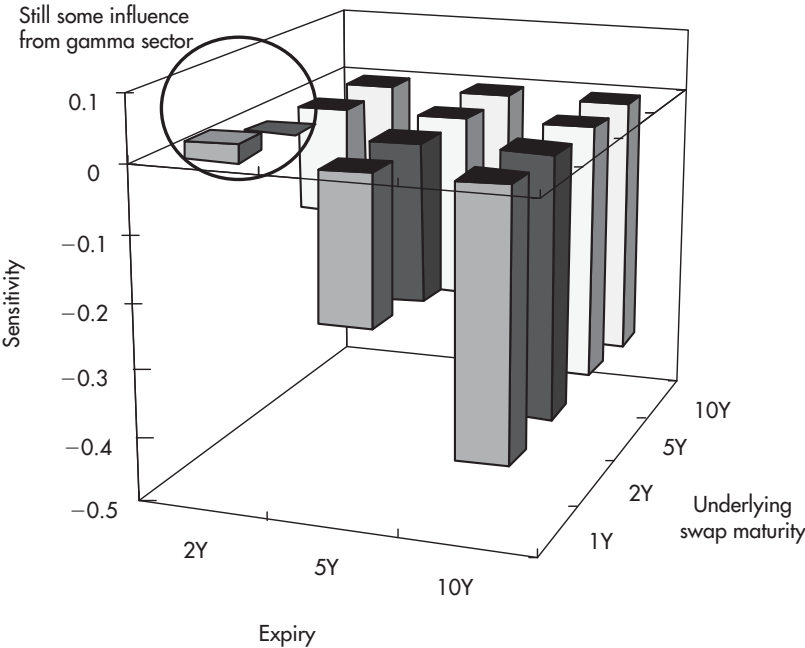


FIGURE 19.21 First eigenvector of a PCA on the vega sector of the JPY volatility surface.
Sources: data – Bloomberg; chart – Authors.
Data period: 5 Jan 2009 to 19 Sep 2011, weekly data.

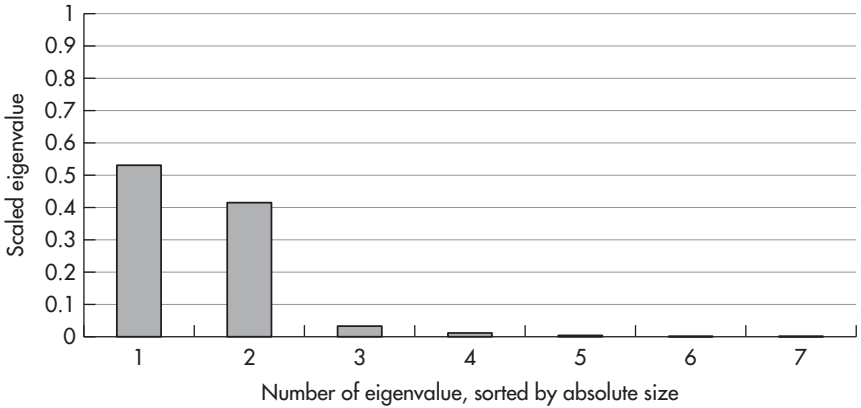


FIGURE 19.22 Scaled eigenvalues of a PCA on the vega sector of the JPY volatility surface.
Sources: data – Bloomberg; chart – Authors.
Data period: 5 Jan 2009 to 19 Sep 2011, weekly data.

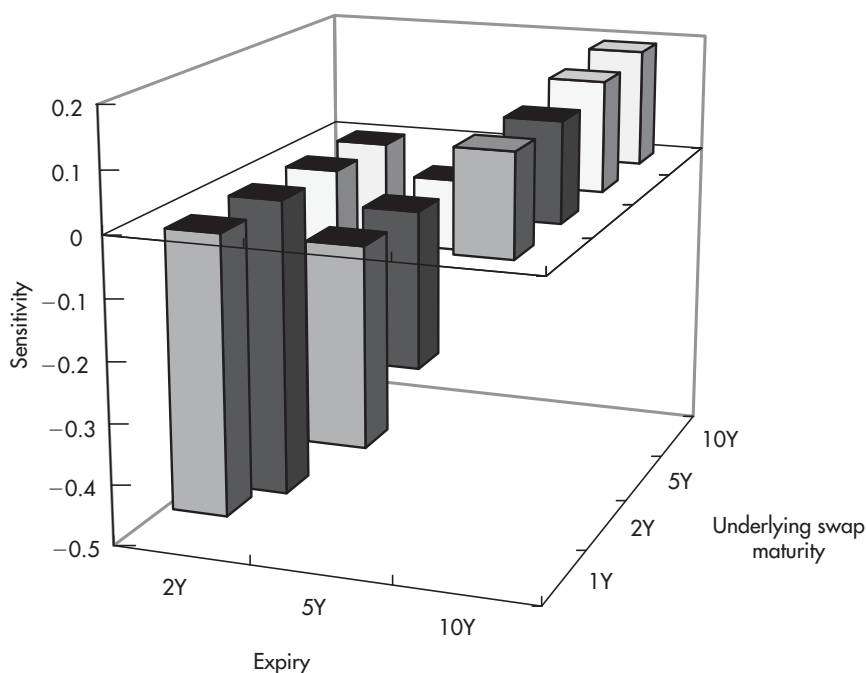


FIGURE 19.23 Second eigenvector of a PCA on the vega sector of the JPY volatility surface.

Sources: data – Bloomberg; chart – Authors.

Data period: 5 Jan 2009 to 19 Sep 2011, weekly data.

of the volatility surface and is not already explained by the overall volatility level unspecific to a certain dimension. The result is an appealing breakdown of the pricing action among the multitude of options into three factors and two dimensions, with one overall factor (1) and one dimension-specific factor (2 and 3) for each of the two dimensions. This result can also serve as a basis for trades on both implied volatility curves (those with variation across expiry and those with variation across underlying swap maturity). As explained above, trades on factor 2 (i.e. over different expiries) are only reasonable in the vega part of the volatility surface.¹⁸ In that segment, however, the high scaled second eigenvalue (Figure 19.22) suggests that they have a

¹⁸Correspondingly, trades in the gamma sector always used options of the same expiry and considered only relative value across the volatility curve over different underlying swap maturities.

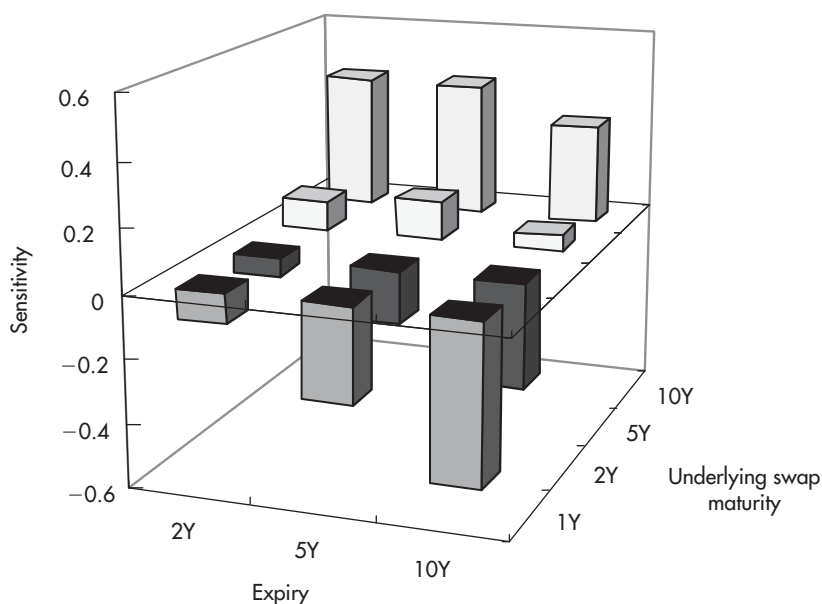


FIGURE 19.24 Third eigenvector of a PCA on the vega sector of the JPY volatility surface.

Sources: data – Bloomberg; chart – Authors.

Data period: 5 Jan 2009 to 19 Sep 2011, weekly data.

significant impact on the shape of the volatility surface. PCA provides the right framework to approach those important trading strategies, and we will give an example subsequently.

These results can vary for other option markets, and it is possible that factors 2 and 3 change place, that is that the variation across the dimension of same-expiry volatility curves is more important (has a higher scaled eigenvalue) than the variation across the dimension of same-underlying-swap-maturity volatility curves. If the second and third eigenvalues are close (unlike in Japan), it is even possible that in the same option market such a switch takes place (i.e. that the two dimensions have different relative importance at different points in time). This is a problem for PCA-based trades on the volatility surface, which we will discuss in the next section.

The statistical properties of the first three factors displayed in Figure 19.25 classify trades both on factors 1 and 2 as rather long-term (slow speed of mean reversion) macroeconomic strategies. Also note the significant correlation between factors 1 and 2 during subperiods, which can be a problem for